Ablation : ConvNeXt-Tiny — GELU | Dropout | BatchNorm

Config: ImageNet- K pretrained | GELU activation | Dropout | BatchNorm **Model:** ConvNeXt-T

Dataset: Rice Leaf Disease Identification Dataset

. Colab Setup (jalankan sekali per session)

In [1]:
```
from google.colab import drive
drive.mount('/content/drive')
import os
DRIVE_DIR = '/content/drive/MyDrive/rice-convnext/04_GELU_Dropout_BatchNo
os.makedirs(DRIVE_DIR, exist_ok=True)
print(f'Drive folder ready: {DRIVE_DIR}')
```

```
Mounted at /content/drive
Drive folder ready: /content/drive/MyDrive/rice-convnext/04_GELU_Dropout_Batc
```

In [2]:
```
from google.colab import files
import os, shutil
print('Upload kaggle.json...')
files.upload()
os.makedirs('/root/.kaggle', exist_ok=True)
shutil.move('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 0o600)
print('Kaggle API ready!')
```

```
Upload kaggle.json...
```

Upload widget is only available when the cell has been executed in the current browser session. Pl
rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
Kaggle API ready!
```

In [3]:
```
DATASET_DIR = '/content/drive/MyDrive/rice-convnext/dataset'
if os.path.exists(DATASET_DIR) and len(os.listdir(DATASET_DIR)) > 0:
    print('Dataset sudah ada, skip download.')
else:
    !pip install kaggle -q
    !kaggle datasets download -d wangxiaoqii/rice-leaf-disease-identifica
    !unzip -q /content/tmp/*.zip -d {DATASET_DIR}
    !rm -rf /content/tmp
    print('Dataset ready!')

import shutil
LOCAL_DATASET = '/content/dataset'
if not os.path.exists(LOCAL_DATASET):
    print('Copying dataset ke local SSD...')
    shutil.copytree(DATASET_DIR, LOCAL_DATASET)
    print('Done!')
else:
```

```
    print('Sudah ada di local.')
DATA_DIR = LOCAL_DATASET
```

```
Dataset sudah ada, skip download.
Copying dataset ke local SSD...
Done!
```

. Setup & Imports

In [4]:
```
import os, random, shutil
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
from sklearn.metrics import classification_report, confusion_matrix
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms, models

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
torch.backends.cudnn.deterministic = True

DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {DEVICE}')
if torch.cuda.is_available():
    print(f'GPU: {torch.cuda.get_device_name(0)}')
```

```
Device: cuda
GPU: Tesla T4
```

. Configuration

In [5]:
```
DATA_DIR      = '/content/dataset'
DRIVE_DIR     = '/content/drive/MyDrive/rice-convnext/04_GELU_Dropout_Bat
IMG_SIZE      = 224
BATCH_SIZE    = 64
NUM_EPOCHS    = 30
LR_INIT       = 1e-3
LR_FINAL      = 1e-6
WEIGHT_DECAY  = 0.05
NUM_WORKERS   = 2
IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD  = [0.229, 0.224, 0.225]
SAVE_PATH     = f'{DRIVE_DIR}/convnext_tiny_04_gelu_dropout_batchnorm.pt'
EXP_NAME      = 'ConvNeXt-Tiny | GELU | Dropout | BatchNorm'
print(f'Experiment: {EXP_NAME}')
```

```
Experiment: ConvNeXt-Tiny | GELU | Dropout | BatchNorm
```

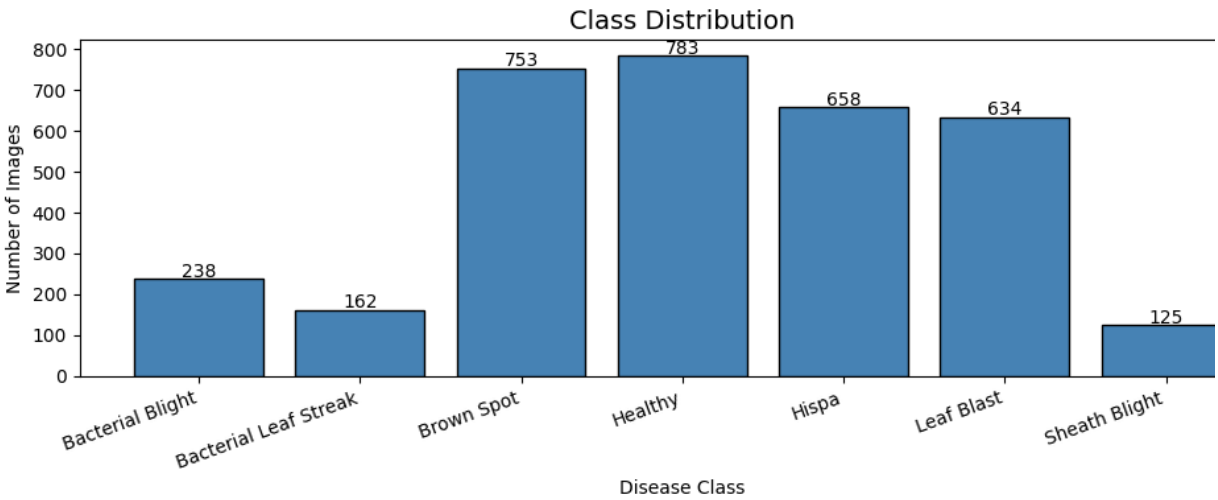
. Data Loading & Augmentation

In [6]:
```
train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])
val_test_transforms = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(IMG_SIZE),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])

full_dataset = datasets.ImageFolder(DATA_DIR)
CLASS_NAMES  = full_dataset.classes
NUM_CLASSES  = len(CLASS_NAMES)
print(f'Classes ({NUM_CLASSES}): {CLASS_NAMES}')
print(f'Total images: {len(full_dataset)}')

total   = len(full_dataset)
n_train = int(0.6 * total)
n_val   = int(0.2 * total)
n_test  = total - n_train - n_val

generator = torch.Generator().manual_seed(SEED)
train_set, val_set, test_set = random_split(
    full_dataset, [n_train, n_val, n_test], generator=generator
)

train_set.dataset.transform = train_transforms
val_set.dataset.transform   = val_test_transforms
test_set.dataset.transform  = val_test_transforms

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False
test_loader  = DataLoader(test_set,  batch_size=BATCH_SIZE, shuffle=False
print(f'Train: {len(train_set)} | Val: {len(val_set)} | Test: {len(test_s
```

```
Classes (7): ['Bacterial Blight', 'Bacterial Leaf Streak', 'Brown Spot', 'Hea
'Hispa', 'Leaf Blast', 'Sheath Blight']
Total images: 3353
Train: 2011 | Val: 670 | Test: 672
```

In [7]:
```
labels_all = [full_dataset.targets[i] for i in range(len(full_dataset))]
counts = [Counter(labels_all)[i] for i in range(NUM_CLASSES)]
plt.figure(figsize=(10, 4))
bars = plt.bar(CLASS_NAMES, counts, color='steelblue', edgecolor='black')
plt.title('Class Distribution', fontsize=14)
plt.xlabel('Disease Class'); plt.ylabel('Number of Images')
plt.xticks(rotation=20, ha='right')
for bar, count in zip(bars, counts):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+5, str(cou
plt.tight_layout()
```

```
plt.savefig('class_distribution.png', dpi=150)
plt.show()
```



. Model — ConvNeXt-Tiny (GELU | Dropout | BatchN

In [8]:
```
class BatchNormNHWC(nn.Module):
    """BatchNorm2d wrapper that accepts NHWC input (used inside ConvNeXt
    def __init__(self, num_features):
        super().__init__()
        self.bn = nn.BatchNorm2d(num_features)
    def forward(self, x):  # x: (B, H, W, C)
        return self.bn(x.permute(0, 3, 1, 2)).permute(0, 2, 3, 1)

def replace_layernorm_with_batchnorm(module):
    """Recursively replace all LayerNorm variants with BatchNorm."""
    for name, child in module.named_children():
        if type(child).__name__ == 'LayerNorm2d':
            # Stem / downsampling: NCHW input → use BatchNorm2d
            C = child.normalized_shape[0]
            setattr(module, name, nn.BatchNorm2d(C))
        elif isinstance(child, nn.LayerNorm):
            # CNBlock (NHWC after Permute) → use BatchNormNHWC
            C = child.normalized_shape[0]
            setattr(module, name, BatchNormNHWC(C))
        else:
            replace_layernorm_with_batchnorm(child)

from torchvision.ops import StochasticDepth

def replace_stochastic_depth(model, p=0.1):
    for name, module in model.named_children():
        if isinstance(module, StochasticDepth):
            setattr(model, name, nn.Dropout(p=p))
        else:
            replace_stochastic_depth(module, p)

model = models.convnext_tiny(weights=models.ConvNeXt_Tiny_Weights.IMAGENE
in_features = model.classifier[2].in_features
model.classifier[2] = nn.Linear(in_features, NUM_CLASSES)

# Replace Stochastic Depth → Dropout(p=0.1)
replace_stochastic_depth(model, p=0.1)
print('Regularization: StochasticDepth → Dropout(p=0.1)')
```

```
# Replace LayerNorm → BatchNorm
replace_layernorm_with_batchnorm(model)
# Fix classifier: after Flatten input is (B, C), use BatchNorm1d
if isinstance(model.classifier[2], BatchNormNHWC):
    C = model.classifier[2].bn.num_features
    model.classifier[2] = nn.BatchNorm1d(C)
print('Normalization: LayerNorm → BatchNorm')

model = model.to(DEVICE)
total_p     = sum(p.numel() for p in model.parameters())
trainable_p = sum(p.numel() for p in model.parameters() if p.requires_gra
print(f'Experiment : {EXP_NAME}')
print(f'Total params    : {total_p:,}')
print(f'Trainable params: {trainable_p:,}')
```

```
Downloading: "https://download.pytorch.org/models/convnext_tiny-983f1562.pth"
root/.cache/torch/hub/checkpoints/convnext_tiny-983f1562.pth
100%|██████████| 109M/109M [00:00<00:00, 118MB/s]
Regularization: StochasticDepth → Dropout(p=0.1)
Normalization: LayerNorm → BatchNorm
Experiment : ConvNeXt-Tiny | GELU | Dropout | BatchNorm
Total params    : 27,825,511
Trainable params: 27,825,511
```

. Training Setup

In [9]:
```
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=LR_INIT, weight_decay=WEIG

def get_scheduler(optimizer, num_epochs, lr_init, lr_final):
    warmup = optim.lr_scheduler.LinearLR(optimizer, start_factor=0.01, en
    cosine = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=num_ep
    return optim.lr_scheduler.SequentialLR(optimizer, schedulers=[warmup,

scheduler = get_scheduler(optimizer, NUM_EPOCHS, LR_INIT, LR_FINAL)
```

. Training Loop

In [10]:
```
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for images, labels in loader:
        images, labels = images.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * images.size(0)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total   += labels.size(0)
    return running_loss / total, correct / total

def evaluate(model, loader, criterion):
    model.eval()
```

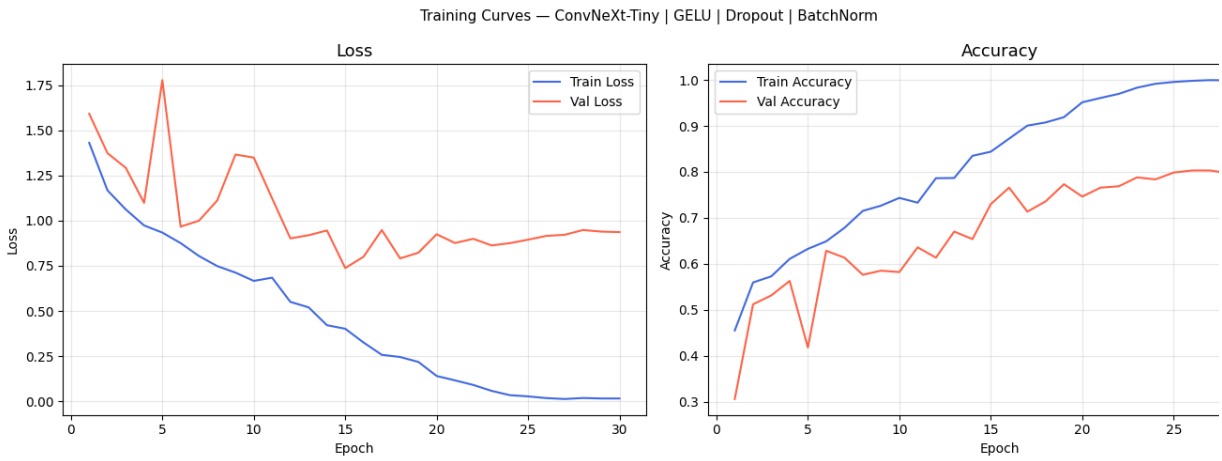
```
    running_loss, correct, total = 0.0, 0, 0
    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = outputs.max(1)
            correct += predicted.eq(labels).sum().item()
            total   += labels.size(0)
    return running_loss / total, correct / total

history = {'train_loss': [], 'train_acc': [], 'val_loss': [], 'val_acc':
best_val_acc = 0.0

for epoch in range(NUM_EPOCHS):
    train_loss, train_acc = train_one_epoch(model, train_loader, criterio
    val_loss,   val_acc   = evaluate(model, val_loader, criterion)
    scheduler.step()
    history['train_loss'].append(train_loss)
    history['train_acc'].append(train_acc)
    history['val_loss'].append(val_loss)
    history['val_acc'].append(val_acc)
    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), SAVE_PATH)
    print(f'Epoch [{epoch+1:02d}/{NUM_EPOCHS}] '
          f'Train Loss: {train_loss:.4f} Acc: {train_acc:.4f} | '
          f'Val Loss: {val_loss:.4f} Acc: {val_acc:.4f}'
          + (' <- best' if val_acc == best_val_acc else ''))

print(f'\nBest Val Accuracy: {best_val_acc:.4f}')
print(f'Saved: {SAVE_PATH}')
```

```
Epoch [01/30] Train Loss: 1.4306 Acc: 0.4550 | Val Loss: 1.5916 Acc: 0.3060 <
Epoch [02/30] Train Loss: 1.1675 Acc: 0.5594 | Val Loss: 1.3734 Acc: 0.5119 <
Epoch [03/30] Train Loss: 1.0620 Acc: 0.5728 | Val Loss: 1.2924 Acc: 0.5313 <
Epoch [04/30] Train Loss: 0.9735 Acc: 0.6106 | Val Loss: 1.0984 Acc: 0.5627 <
Epoch [05/30] Train Loss: 0.9339 Acc: 0.6325 | Val Loss: 1.7776 Acc: 0.4179
Epoch [06/30] Train Loss: 0.8757 Acc: 0.6489 | Val Loss: 0.9669 Acc: 0.6284 <
Epoch [07/30] Train Loss: 0.8045 Acc: 0.6783 | Val Loss: 0.9996 Acc: 0.6134
Epoch [08/30] Train Loss: 0.7485 Acc: 0.7151 | Val Loss: 1.1118 Acc: 0.5761
Epoch [09/30] Train Loss: 0.7125 Acc: 0.7265 | Val Loss: 1.3661 Acc: 0.5851
Epoch [10/30] Train Loss: 0.6667 Acc: 0.7434 | Val Loss: 1.3488 Acc: 0.5821
Epoch [11/30] Train Loss: 0.6845 Acc: 0.7330 | Val Loss: 1.1236 Acc: 0.6358 <
Epoch [12/30] Train Loss: 0.5506 Acc: 0.7862 | Val Loss: 0.9018 Acc: 0.6134
Epoch [13/30] Train Loss: 0.5202 Acc: 0.7867 | Val Loss: 0.9195 Acc: 0.6701 <
Epoch [14/30] Train Loss: 0.4216 Acc: 0.8349 | Val Loss: 0.9457 Acc: 0.6537
Epoch [15/30] Train Loss: 0.4018 Acc: 0.8439 | Val Loss: 0.7376 Acc: 0.7299 <
Epoch [16/30] Train Loss: 0.3258 Acc: 0.8722 | Val Loss: 0.8004 Acc: 0.7657 <
Epoch [17/30] Train Loss: 0.2577 Acc: 0.9005 | Val Loss: 0.9482 Acc: 0.7134
Epoch [18/30] Train Loss: 0.2455 Acc: 0.9075 | Val Loss: 0.7912 Acc: 0.7358
Epoch [19/30] Train Loss: 0.2182 Acc: 0.9189 | Val Loss: 0.8227 Acc: 0.7731 <
Epoch [20/30] Train Loss: 0.1405 Acc: 0.9513 | Val Loss: 0.9248 Acc: 0.7463
Epoch [21/30] Train Loss: 0.1163 Acc: 0.9607 | Val Loss: 0.8763 Acc: 0.7657
Epoch [22/30] Train Loss: 0.0910 Acc: 0.9697 | Val Loss: 0.8997 Acc: 0.7687
Epoch [23/30] Train Loss: 0.0582 Acc: 0.9831 | Val Loss: 0.8630 Acc: 0.7881 <
Epoch [24/30] Train Loss: 0.0343 Acc: 0.9915 | Val Loss: 0.8756 Acc: 0.7836
Epoch [25/30] Train Loss: 0.0277 Acc: 0.9955 | Val Loss: 0.8949 Acc: 0.7985 <
Epoch [26/30] Train Loss: 0.0180 Acc: 0.9980 | Val Loss: 0.9158 Acc: 0.8030 <
Epoch [27/30] Train Loss: 0.0131 Acc: 0.9995 | Val Loss: 0.9218 Acc: 0.8030 <
Epoch [28/30] Train Loss: 0.0186 Acc: 0.9990 | Val Loss: 0.9482 Acc: 0.7970
Epoch [29/30] Train Loss: 0.0160 Acc: 0.9985 | Val Loss: 0.9395 Acc: 0.8000
Epoch [30/30] Train Loss: 0.0159 Acc: 0.9975 | Val Loss: 0.9367 Acc: 0.7985

Best Val Accuracy: 0.8030
Saved: /content/drive/MyDrive/rice-convnext/04_GELU_Dropout_BatchNorm/
convnext_tiny_04_gelu_dropout_batchnorm.pt
```

. Training Curves

In [11]:

```
epochs_range = range(1, NUM_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(epochs_range, history['train_loss'], label='Train Loss', col
axes[0].plot(epochs_range, history['val_loss'],   label='Val Loss',   col
axes[0].set_title('Loss', fontsize=13)
axes[0].set_xlabel('Epoch'); axes[0].set_ylabel('Loss')
axes[0].legend(); axes[0].grid(alpha=0.3)
axes[1].plot(epochs_range, history['train_acc'], label='Train Accuracy',
axes[1].plot(epochs_range, history['val_acc'],   label='Val Accuracy',
axes[1].set_title('Accuracy', fontsize=13)
axes[1].set_xlabel('Epoch'); axes[1].set_ylabel('Accuracy')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.suptitle(f'Training Curves — {EXP_NAME}', fontsize=11)
plt.tight_layout()
plt.savefig('training_curves.png', dpi=150)
plt.show()
```



. Evaluation on Test Set

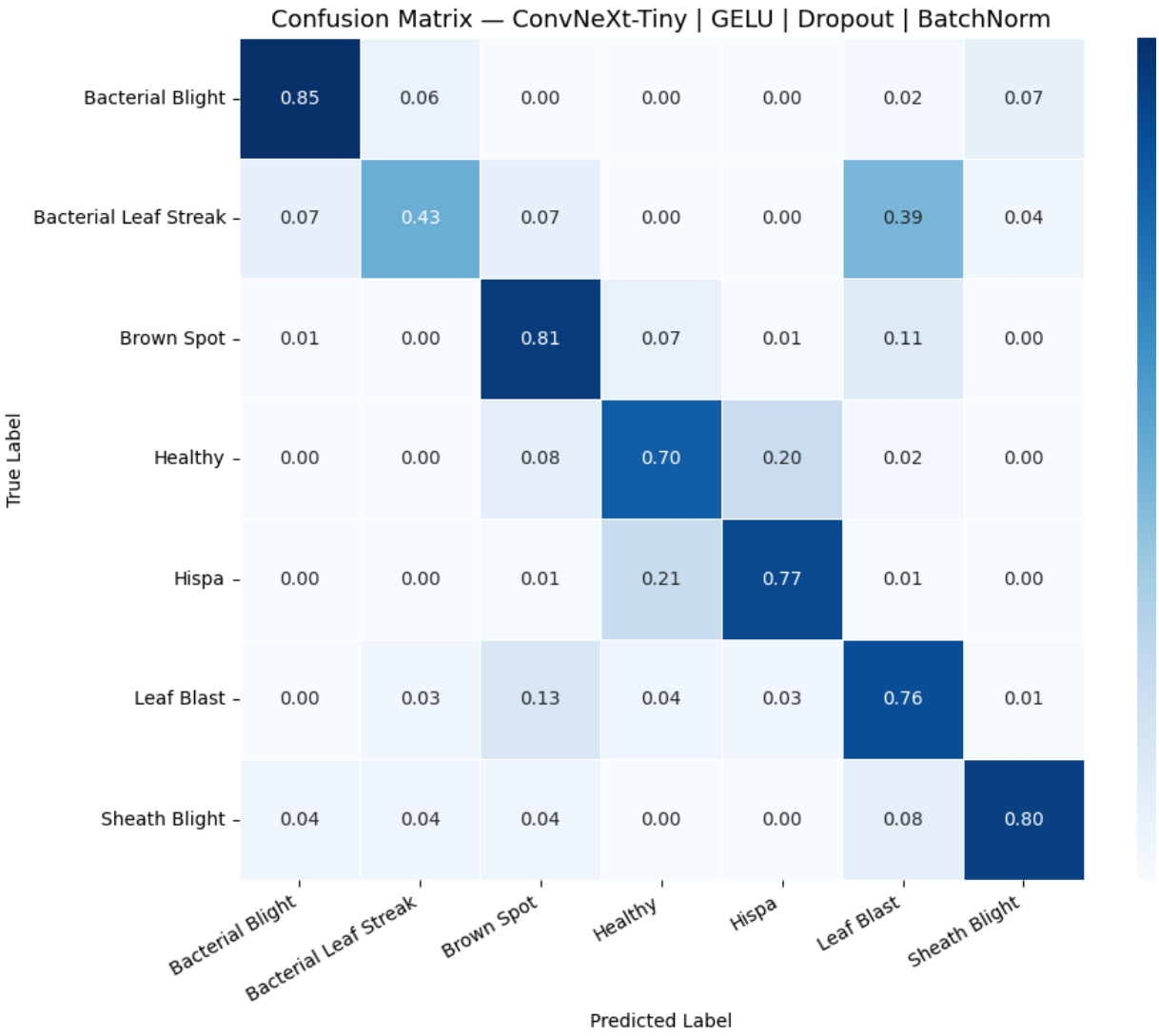
In [12]:
```
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
all_preds, all_labels = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(DEVICE)
        outputs = model(images)
        _, predicted = outputs.max(1)
        all_preds.extend(predicted.cpu().numpy())
        all_labels.extend(labels.numpy())

report = classification_report(all_labels, all_preds, target_names=CLASS_
report_df = pd.DataFrame(report).transpose()
print('=== Classification Report ===')
print(report_df.round(4).to_string())
report_df.to_csv('classification_report.csv')

summary = {
    'Model'     : ['ConvNeXt-Tiny'],
    'Activation': ['GELU'],
    'Dropout'   : ['Dropout'],
    'Norm'      : ['BatchNorm'],
    'Accuracy'  : [report['accuracy']],
    'Precision' : [report['weighted avg']['precision']],
    'Recall'    : [report['weighted avg']['recall']],
    'F1-Score'  : [report['weighted avg']['f1-score']],
}
summary_df = pd.DataFrame(summary)
print('\n=== Summary ===')
print(summary_df.round(4).to_string(index=False))
summary_df.to_csv('summary.csv', index=False)
```

```
=== Classification Report ===
                      precision  recall  f1-score   support
Bacterial Blight         0.9200  0.8519    0.8846   54.0000
Bacterial Leaf Streak    0.6000  0.4286    0.5000   28.0000
Brown Spot               0.8036  0.8133    0.8084  166.0000
Healthy                  0.7484  0.6988    0.7227  166.0000
Hispa                    0.6911  0.7727    0.7296  110.0000
Leaf Blast               0.7154  0.7561    0.7352  123.0000
Sheath Blight            0.7692  0.8000    0.7843   25.0000
accuracy                 0.7545  0.7545    0.7545    0.7545
macro avg                0.7497  0.7316    0.7378  672.0000
weighted avg             0.7550  0.7545    0.7533  672.0000

=== Summary ===
        Model Activation Dropout      Norm  Accuracy  Precision  Recall  F1-S
ConvNeXt-Tiny       GELU Dropout BatchNorm    0.7545      0.755  0.7545    0.
```

. Confusion Matrix

In [13]:

```python
cm = confusion_matrix(all_labels, all_preds)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
plt.figure(figsize=(10, 8))
sns.heatmap(cm_norm, annot=True, fmt='.2f',
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES,
            cmap='Blues', linewidths=0.5)
plt.title(f'Confusion Matrix — ConvNeXt-Tiny | GELU | Dropout | BatchNorm
plt.ylabel('True Label'); plt.xlabel('Predicted Label')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=150)
plt.show()
```



. Save All to Drive

In [14]:
```
local_files = ['training_curves.png', 'confusion_matrix.png',
               'classification_report.csv', 'summary.csv', 'class_distrib
for fname in local_files:
    if os.path.exists(fname):
        shutil.copy(fname, f'{DRIVE_DIR}/{fname}')

all_outputs = [
    (f'{DRIVE_DIR}/convnext_tiny_04_gelu_dropout_batchnorm.pt',
    (f'{DRIVE_DIR}/training_curves.png',              'Training curves'),
    (f'{DRIVE_DIR}/confusion_matrix.png',             'Confusion matrix')
    (f'{DRIVE_DIR}/classification_report.csv',        'Per-class metrics'
    (f'{DRIVE_DIR}/summary.csv',                      'Summary table'),
]
print(f'=== Output Files: {DRIVE_DIR} ===')
for fpath, desc in all_outputs:
    exists = '✅' if os.path.exists(fpath) else '❌'
    print(f'{exists}  {os.path.basename(fpath):<45} {desc}')
```

```
=== Output Files: /content/drive/MyDrive/rice-convnext/04_GELU_Dropout_BatchN
===
✅  convnext_tiny_04_gelu_dropout_batchnorm.pt    Model weights
✅  training_curves.png                           Training curves
✅  confusion_matrix.png                          Confusion matrix
✅  classification_report.csv                     Per-class metrics
✅  summary.csv                                   Summary table
```